03. Emma and triangles

5 marks

Problem statement

Emma is drawing an abstract line art represented as a complete graph of **N** nodes made of blue and red lines. You will be given **M** red-colored edges as the input and the rest of the edges will be blue-colored. Find out the total number of unique single-colored triangles in the graph. If the answer exceeds $10^9 + 7$, print the answer modulo (%) $10^9 + 7$.

Input Format:

The input is in the following format:

- The first line contains two space-separated integers denoting **N** and **M** respectively.
- The next **M** lines contain 2 space-separated integers denoting two different nodes connected with a red-colored edge.

The input will be read from the STDIN by the candidate

Output Format:

The output contains an integer that denotes the total number of unique single-colored triangles.

The output will be matched to the candidate's output printed on the STDOUT

Constraints:

- $3 \leq N \leq 10^5$
- $1 \leq M \leq 10^5$

Example:

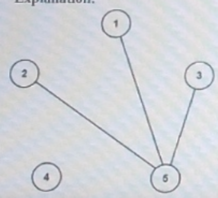
Input:

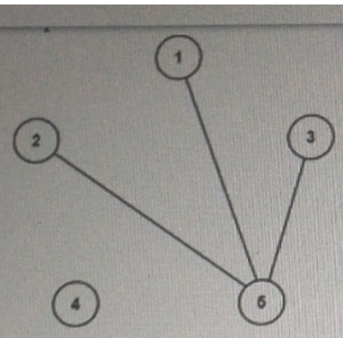
5 3
1 5
2 5
3 5

Output:

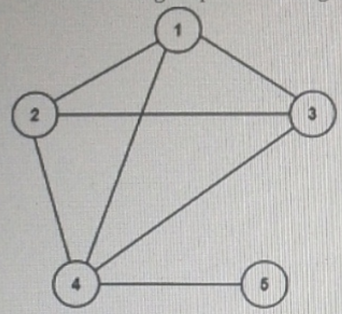
4

Explanation:





The above image represents the graph with only red edges. No triangles are formed with the red edges.



The above image represents the graph with only blue edges. Four triangles are formed with the blue edges.
Following are the list of triangles with a single color:
(1,2,3), (1,2,4), (2,3,4), (1,3,4)
So, the output is 4 (0+4).

Sample Input

```
4 3
1 2
2 4
4 3
```

Sample Output

```
0
```

Instructions :

- Program should take input from standard input and print output to standard output.
- Your code is judged by an automated system, do not write any additional welcome/greeting messages.
- "Save and Test" only checks for basic test cases, more rigorous cases will be used to judge your code while scoring.
- Additional score will be given for writing optimized code both in terms of memory and execution time.

Now let's start coding :